

Lean Multi-Agent Systems on Azure

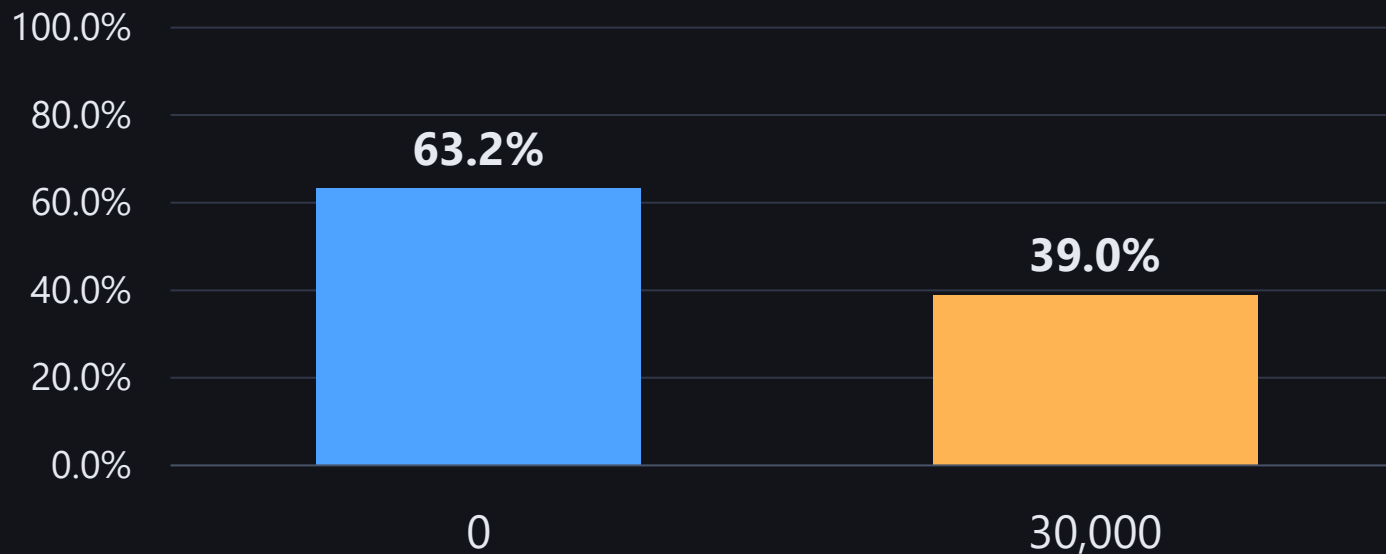
Microsoft Agent Framework
with Azure OpenAI



The facts can be present and the answer still wrong

More irrelevant text inside the prompt. Same question and evidence.

Answer accuracy: Llama 3 8B on MMLU questions



Additional irrelevant tokens inside the model input

Accuracy fell 24.2 percentage points.

Du et al., October 2025 preprint, Table 6. Controlled benchmark with an older 8B model.

Copying the evidence

97%

At both lengths, on a separate exact-copy test.

Finding the facts does not guarantee a correct answer.

Useful boundaries for a team

Useful boundaries

Distinct domain instructions and tools

Focused assignments with limited context

Independent work whose results combine

Clear coordination

A named owner for the customer reply

Explicit prerequisites for ordered work

Checks against the original request

Every boundary should preserve what the next agent needs.

Measure the complete outcome, including coordination calls and failed attempts.

The workload changes the result

Relative task performance: centralized agents vs a single agent

Finance Agent

+80.8%

Financial-analysis benchmark
Earnings, ratios, market trends

PlanCraft

−50.3%

Minecraft crafting benchmark
Ingredients and actions in order

The same coordination design can help one workload and hurt another.

Kim et al., §4.2. December 2025 paper, revised April 2026. Relative performance changes, not token savings.

Alex needs help before the next hike

Defective boots, an upcoming trip, and unfinished work

Return

Defective boots from order #48213

Replacement recommendation

Under \$150, in stock in size 10

Address update

42 Summit Trail

Same starting request across alternative designs. Fictional local business data.



A multi-agent team for Alex

Now we divide Alex's request among four specialists

Orders

Order details and
shipment evidence

Returns

Defect eligibility and
the return action

Product

Price, size, and
available stock

Account

Customer profile
and address
changes

Coordinator roles change by pattern. Demo 5 uses one concierge.

Patterns for sharing the work

A different choice about assignment, ownership, or participation

Scoped delegation

Supervisor assigns tasks and combines replies.

Fits: One request spans specialties.

Risk: A brief drops a constraint or an action.

Handoff

A specialist takes over the conversation.

Fits: The next domain should own the reply.

Risk: Facts or ownership get lost in transfer.

Routing

Select a specialist or workflow.

Fits: Requests fall into clear categories.

Risk: A vague follow-up takes the wrong path.

These patterns combine: our router sends a multi-domain request to the supervisor.

Patterns for execution and memory

Dependencies, current conversation, and the next visit

Plan and execute

Declare dependencies, then schedule tasks.

Fits: Variable work with prerequisites.

Risk: A valid plan misses a business condition.

Rolling summary

Keep recent turns and summarize older discussion.

Fits: A long conversation in one session.

Risk: A correction or uncertainty disappears.

Durable facts

Save and retrieve selected facts across visits.

Fits: A returning customer needs continuity.

Risk: A saved preference is wrong or stale.

Memory also helps one agent. Every pattern still needs an outcome check.

A bigger business, a focused assignment

The business grows. Alex still needs boots.

Alex's request

Return + replacement recommendation
+ address update

The wider business

More departments
More policies and tools

Product needs

Under \$150

Size 10

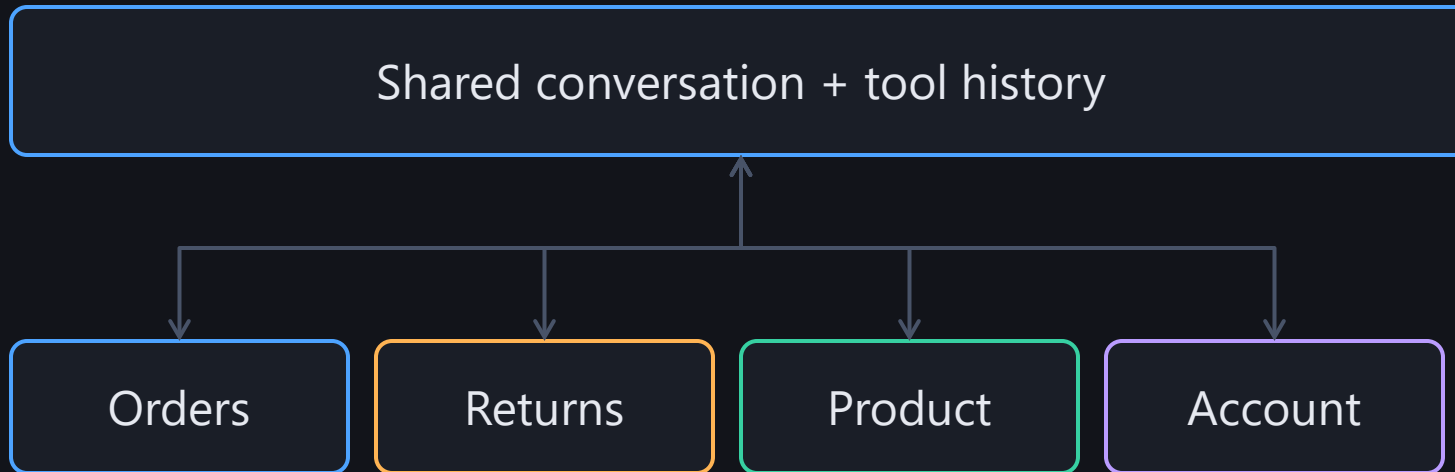
In stock

At 48 available tools, the scoped team used about 53% fewer tokens.

Versus the shared-history team. 4/4 and 4/4 passed. Microsoft Agent Framework.

Demo 0: Shared history can become shared overhead

Every specialist inherits the conversation and earlier tool results



Every specialist takes a turn · Three domain tools each

The next task

Change an address

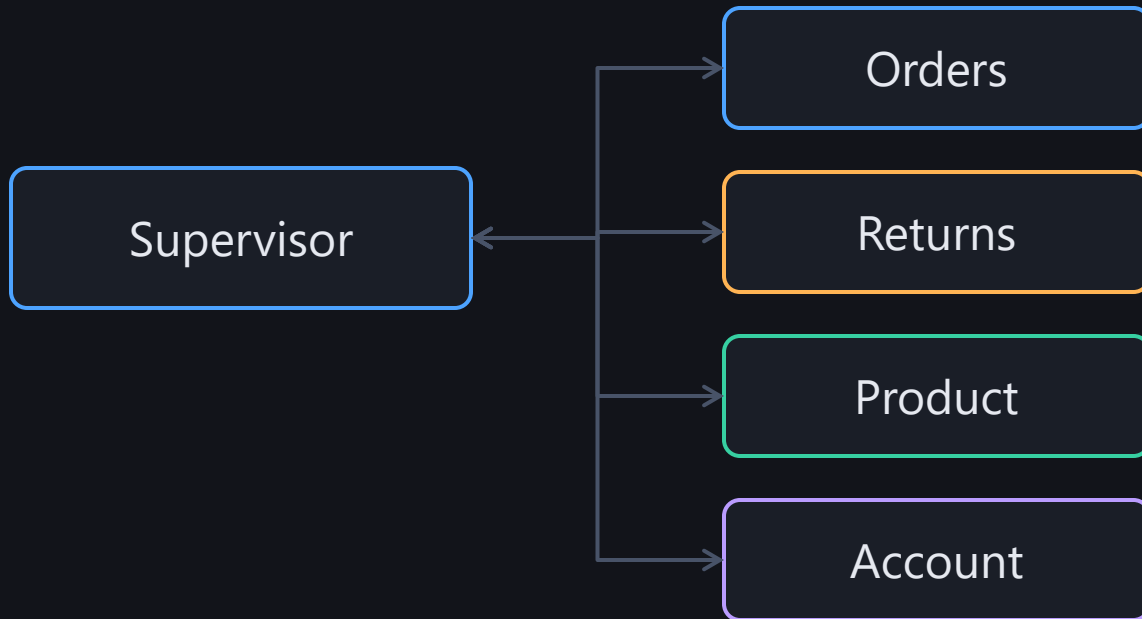
The extra baggage

An entire product catalog

Illustrative setup. Demo 0 shows the actual information reaching Account.

A supervisor delegates focused tasks

Useful when one request crosses specialties and needs one complete reply



Out: task brief
Back: specialist result

Fresh session, standing instructions, domain tools

Where it can fall short

A short brief can still omit the constraint that matters.

Demo 1: did the constraints survive?

Captured scoped brief: fixed code, September 11, round 1

Shared-history input

Standing instructions + tools

Conversation and retained results,
including unrelated work

Focused specialist input

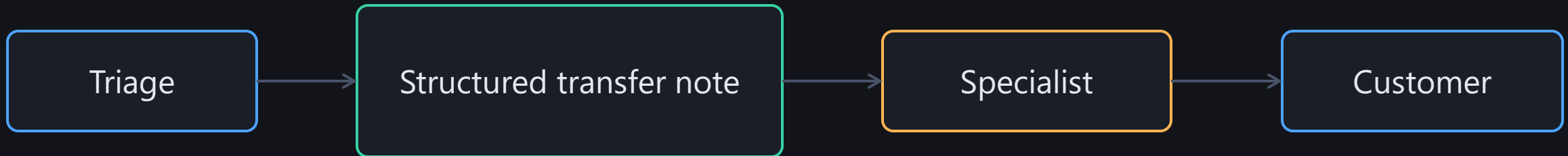
Standing instructions + tools

**“Recommend a replacement for
hiking boots under \$150 that is in
stock in size 10.”**

The brief is requested to be short; its length is not enforced. Each specialist can add its own tool results.

A handoff passes responsibility

Useful when a specialist should take over the customer conversation



Intent, relevant facts, open question

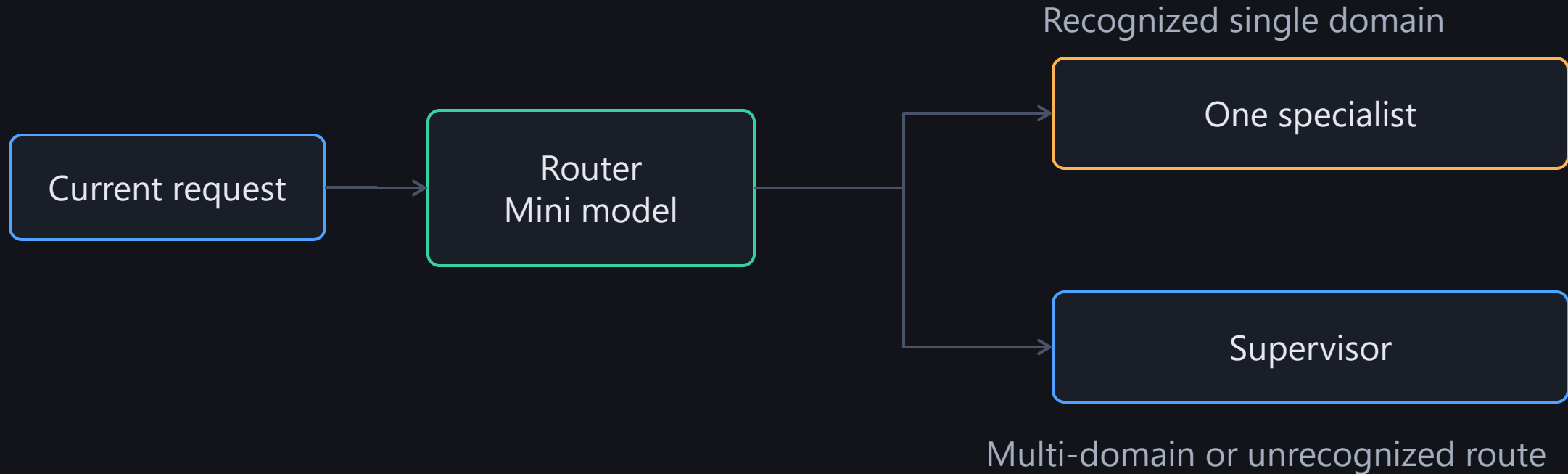
Order #48213 + reported defect = the right return policy

Where it can fall short

Validation rejects empty fields; the handoff can still lose meaning.

A router chooses the path

Useful for distinct request categories. A model classifies and code dispatches.

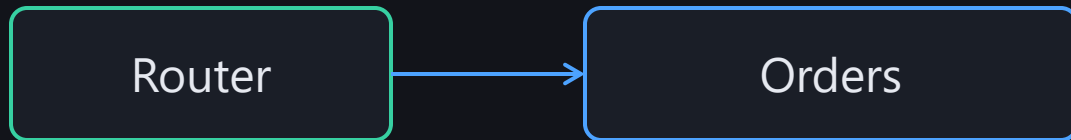


Where it can fall short

A valid route label can still be the wrong choice.

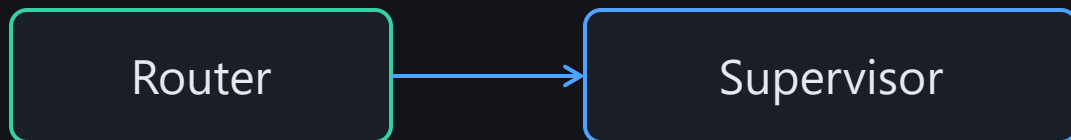
Demo 3: does this request need a team?

“Where’s my order #48610?”



One agent answers

Return + replacement + address

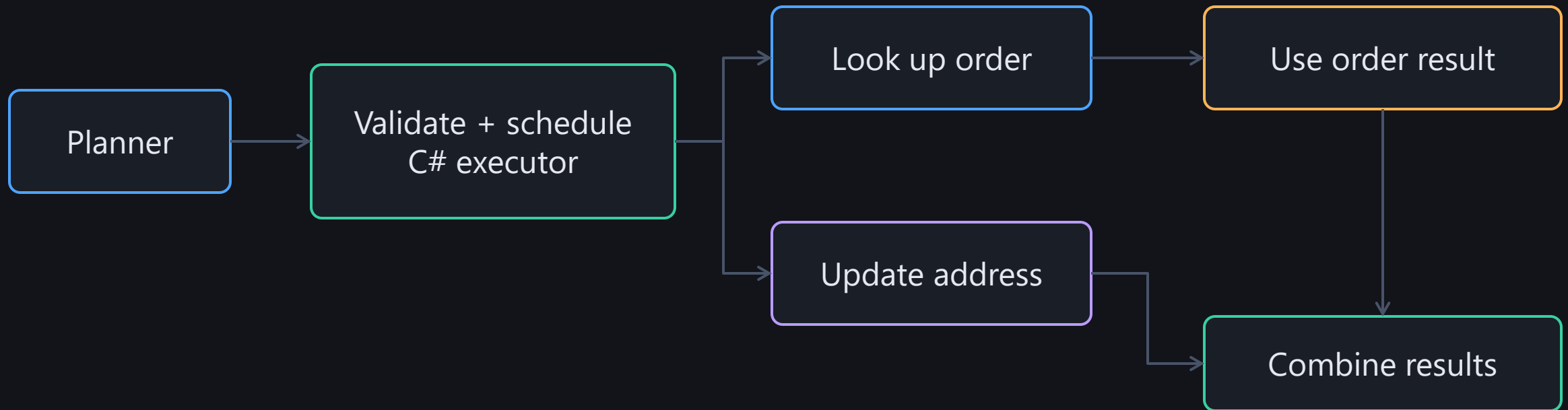


Supervisor splits it up

The router is a model call too. Count it.

A plan makes dependencies visible

Useful when the required tasks vary and some need earlier results

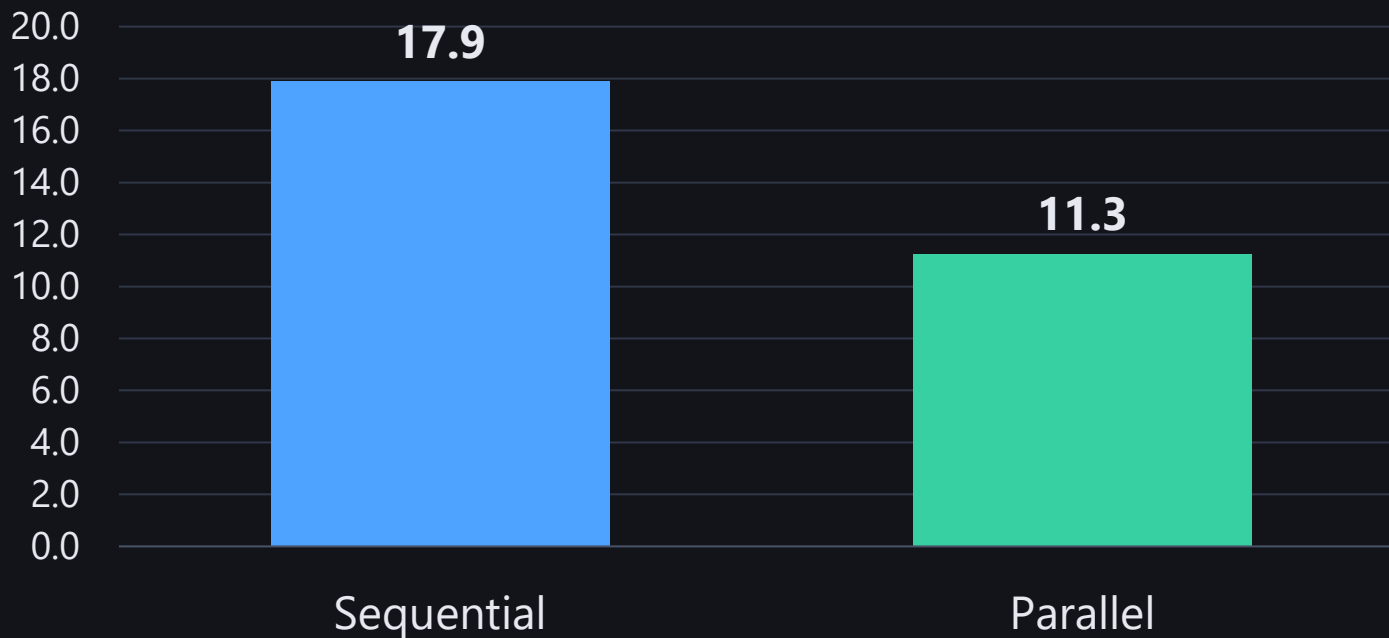


Risk: missing prerequisites or a completed step with the wrong result.

Known steps can use a fixed workflow. This executor waits for each entire batch.

Demo 4: what can run together?

The same saved plan, executed with Microsoft Agent Framework agents



Independent work

Find replacements

Update the address

Returns still checks eligibility before starting the return.

Seconds: workers plus final reply

4 matched pairs. Outcomes: sequential 4/4, parallel 4/4. Planning cost and full results in backup.

Memory has two different jobs

Working context

Continue this conversation

Durable facts

Support the next visit

Research preprint: Mem0 on LOCOMO, April 2025



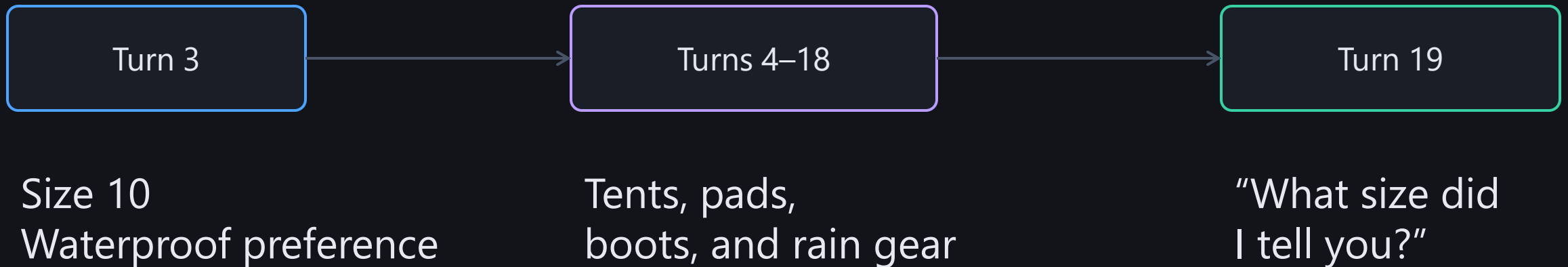
**About 93% fewer
answer-context tokens**

LLM-judge quality score
Full history: 72.90%
Mem0: 66.88%

Chhikara et al., Table 2. Authors' GPT-4o-mini evaluation; context excludes memory-building work.

Demo 5a: what grows with the conversation?

One concierge, the same 20-turn script

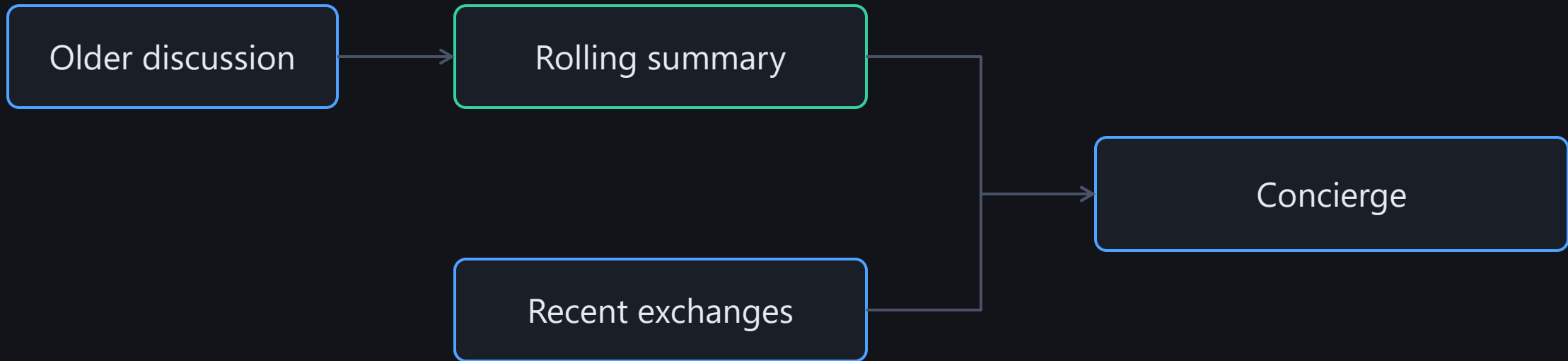


Inspect the early and late input, then check the recalled fact.

Captured script: inspect recall separately from the accuracy of the final packing list.

Demo 5b: did the useful details survive?

Useful for long sessions: a rolling summary plus recent exchanges

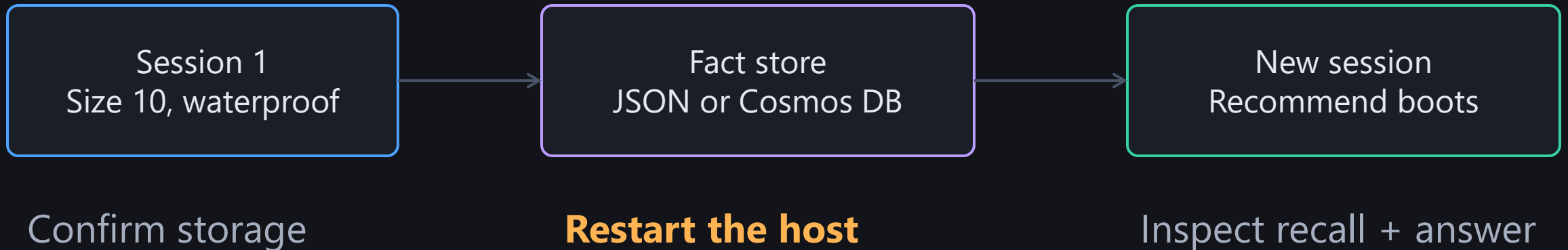


Did size 10 and the waterproof preference survive?

Risk: lost corrections or uncertainty. Count summarization calls and test what survives.

Demo 5c: Alex returns tomorrow

Useful across visits: retrieve selected facts for a new recommendation



Restart recall: durable facts 3/3 · session-only variants 0/3 each

Selected customer messages are stored verbatim. Freshness still needs testing.

Risk: stale or incorrect facts. Test changed preferences. Research: Fan et al., StateMemBench, August 2026.

What happened in our matched runs

September 11 repeat · Three fresh-process attempts per design · Medians

Workload / design	Passed	Calls	Total tokens	Seconds
Request · shared history	3/3	8	23,194	19.9
Request · scoped delegation	2/3	9	10,579	21.7
Handoff · shared history	3/3	8	39,653	19.9
Handoff · structured context	3/3	13	17,471	30.8
One agent · full history	3/3	25	177,281	185.4
One agent · summary	3/3	40	200,021	185.7
One agent · summary + facts	3/3	63	172,609	170.6

Check what finished before comparing token totals.

Memory rows: one concierge, 20 customer turns. Tokens include all model calls; restart probes separate.

DISCOVERAF

Alex came here for boots

Verified example: scoped delegation, September 11, round 1

Return RET-9000 started under the defect policy.

Replacement Four qualifying boots. Size 10. Under \$150.

Address 42 Summit Trail saved in the profile.

Take one real request. Follow every call. Check what happened.

Focused agents, clear handoffs, and useful memory. Measure the complete outcome.